

Mathematics for Computer Science

COMP1021: Durham University

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Part I

Calculus

1 Introduction to Calculus

1.1 Sets

Definition 1.1.1: A set is an unordered collection of distinct elements denoted as $A = \{x | \text{condition}\}$.

Some common set operations include:

- **Definition 1.1.2:** The union of two sets, $A \cup B$, includes all the elements of both sets.
- **Definition 1.1.3:** The intersection of two sets, $A \cap B$, includes any elements which are present in both sets.
- **Definition 1.1.4:** If A and B are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (1)$$

- **Definition 1.1.5:** The set A is a subset of B , $A \subseteq B$, if every element of A is an element of B . A is a proper subset of B , $A \subset B$, if A is a subset of B and there is at least one element of B not in A .
- **Definition 1.1.6:** The powerset of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (2)$$

- **Definition 1.1.7:** The cartesian product of two sets, $A \times B$, is the set of all possible ordered pairs with the first element from A and the second from B .

Definition 1.1.8: For a finite set, the cardinality is the number of distinct elements that it contains.

1.2 Functions

For a function $f : A \rightarrow B$,

- **Definition 1.2.1:** A is known as the domain of f .
- **Definition 1.2.2:** B is known as the codomain of f .
- **Definition 1.2.3:** If $f(a) = b$, then b is the image of a .
- **Definition 1.2.4:** The pre-image of b is the set $\{a | f(a) = b\}$.
- **Definition 1.2.5:** f forms an injection if every element of A maps to a unique element in B , even if some elements of B don't have a mapping.
- **Definition 1.2.6:** f forms a surjection if every element of B maps to at least one element of A .
- **Definition 1.2.7:** f forms a bijection if it is both injective and surjective.

Definition 1.2.8: If for an unbounded set A , there is a bijective function $f : A \rightarrow \mathbb{N}$, then we say A is countably infinite, otherwise A is known as uncountably infinite.

1.3 Hyperbolic Functions

Definition 1.3.1: Given the unit hyperbola $x^2 - y^2 = 1$, the hyperbolic functions are defined by the coordinates of a point at which area $\frac{1}{2}\alpha$ has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

2 Multivariate Functions

2.1 Multivariate Derivatives

2.2 Multivariate Extrema

Part II

Linear Algebra

Part III

References

3 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Calculus* is taught by Doctor Junyan Hu and Doctor Sukanya Pandey.

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- *Linear Algebra* is taught jointly by Professor Steven Bradley and Doctor Karl Southern.

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4 Cited Works